Solution Requirements

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The Flask web application does not require user login. Any user can access the HDI prediction form directly via the browser without authentication.	Low
2	Authorization levels	Single-level access — all users have the same access to the prediction form and results. No admin or restricted roles are defined.	Low
3	External interfaces	The system uses the Kaggle HDI dataset (CSV format) as external data source. The trained model is saved using Pickle and loaded by Flask for inference.	High
4	Transactions processing	User inputs (Life Expectancy, Mean Years of Schooling, GNI per capita, Internet Users) are submitted via HTML POST form. Flask processes and passes them to the ML model for prediction.	High
5	Reporting	The system generates an HDI prediction result displayed on the result page, classifying the input as Low, Medium, High, or Very High HDI along with the numeric score.	High

6	Business rules, etc.	HDI score ranges: 0.3–0.4 = Low HDI, 0.4–0.7 = Medium HDI, 0.7–0.8 = High HDI, 0.8–0.94 = Very High HDI. Values outside this range return an error message.	High
7	Compliance to laws or regulations	The project uses publicly available HDI data from Kaggle. No personal user data is stored. The application complies with basic data usage and open-source licensing requirements.	Medium
8	Other	The system must run on a local Flask server (port 5000) and be accessible via browser. The model file (HDI.pkl) must be present in the Flask folder for predictions to work.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The Flask application must process user inputs and return HDI predictions quickly without noticeable delay.	Prediction response in < 2 seconds on local server
2	Scalability	The current system is designed for academic/local use. It handles single-user requests. Future deployment can scale using cloud hosting.	Supports single concurrent user; scalable to 100+ with deployment
3	Security & Data Privacy	No personal user data is collected or stored. Input values are processed in-memory only. The application runs locally and does not transmit data externally.	No user data stored; local server only; no external data transmission
4	Reliability & Availability	The application must run reliably on the local machine with the Flask development server. The model file must load correctly every time the app starts.	100% uptime during local usage; model loads successfully on every run
5	Usability & Accessibility	The web interface must be simple and easy to use. Input fields are clearly labeled. The prediction result is displayed in plain language understandable to any user.	Any user can complete a prediction in under 1 minute without guidance
6	Other	Maintainability: Code is organized in separate folders (Dataset, Flask, Training). Portability: The project can be run on any system with Python, Flask, and required libraries installed.	Project runs successfully on any OS with Python 3.x and required dependencies